Elementary Analysis

Chapter 5, Sections 28–31

All 53 exercises, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 28–31, 53 exercises. Short descriptions identify the problems; consult the linked second-edition text for the original statements.

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28 Basic properties of the derivative

A derivative measures a limiting slope, not merely a formal manipulation of a formula. At an interior point a,

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

The limit must be a finite real number and must agree from both sides. Differentiability implies continuity; the converse is false. The sum, product, quotient, and chain rules apply only after their hypotheses have been checked. At a point where a formula changes, return to this definition.

28.1 Exercise 28.1: Where absolute values create corners

What we must show. Find all points of nondifferentiability for the six functions.

Plan. Away from zeros of the expression inside an absolute value, its sign stays fixed nearby. Replace the absolute value by either that expression or its negative, and use ordinary differentiation. At a zero, compare the two one-sided slopes.

(a) The function $e^{|x|}$.

For $x > 0$ the function is e^x, and for $x < 0$ it is e^{-x}. Both formulas are differentiable on their respective open half-lines. At zero, the right derivative is 1, whereas the left derivative is -1. The only exceptional point is 0.

(b) The function $\sin|x|$.

On $x > 0$ this is $\sin x$, and on $x < 0$ it is $-\sin x$. The one-sided derivatives at zero are 1 and -1. Again the exceptional set is $\{0\}$.

(c) The function $|\sin x|$.

The possible corners are the zeros $k\pi$, $k \in \mathbb{Z}$. Near $a = k\pi$, write $x = a + h$. Then

$$\frac{|\sin(a+h)| - |\sin a|}{h} = \frac{|\sin h|}{h}.$$

As $h \to 0^+$ this tends to 1; as $h \to 0^-$ it tends to -1. All and only the points $k\pi$ are exceptional.

(d) The function $|x| + |x - 1|$.

Splitting at 0 and 1 gives

$$|x| + |x-1| = \begin{cases} 1 - 2x, & x < 0, \\ 1, & 0 \le x \le 1, \\ 2x - 1, & x > 1. \end{cases}$$

The slopes change from -2 to 0 at 0, and from 0 to 2 at 1. The exceptional set is $\{0, 1\}$.

(e) The function $|x^2 - 1|$.

It equals $x^2 - 1$ for $|x| \ge 1$ and $1 - x^2$ for $|x| < 1$. At -1 the slopes are -2 from the left and 2 from the right; at 1 they are -2 from the left and 2 from the right. Thus the exceptional set is $\{-1, 1\}$.

(f) The function $|x^3 - 8|$.

The inner expression changes sign only at 2. The function is $8 - x^3$ to the left and $x^3 - 8$ to the right, so its slopes at 2 are -12 and 12. Only 2 is exceptional.

Conclusion. In order: $\{0\}$, $\{0\}$, $\{k\pi : k \in \mathbb{Z}\}$, $\{0, 1\}$, $\{-1, 1\}$, and $\{2\}$. The piecewise descriptions also specify the corresponding graph shapes.

28.2 Exercise 28.2: Four derivatives directly from the definition

What we must show. Evaluate the requested limiting difference quotients without using differentiation rules.

(a) The cube at 2.

For $x \neq 2$, factor the difference of cubes:

$$\frac{x^3 - 8}{x - 2} = x^2 + 2x + 4 \longrightarrow 4 + 4 + 4 = 12.$$

Thus $f'(2) = 12$.

(b) The affine function at a.

For $h \neq 0$,

$$\frac{g(a + h) - g(a)}{h} = \frac{a + h + 2 - (a + 2)}{h} = 1.$$

The quotient is already constant, so $g'(a) = 1$ for every real a.

(c) The function $x^2 \cos x$ at zero.

Here $f(0) = 0$, and for $x \neq 0$ the quotient is $x \cos x$. Since $|x \cos x| \leq |x| \to 0$, the squeeze theorem gives $f'(0) = 0$.

(d) The rational function at 1.

For $r(x) = (3x + 4)/(2x - 1)$, we have $r(1) = 7$. In a sufficiently small neighborhood of 1, the denominator is nonzero, and

$$\frac{r(x) - r(1)}{x - 1} = \frac{3x + 4 - 7(2x - 1)}{(2x - 1)(x - 1)} = \frac{-11}{2x - 1} \longrightarrow -11.$$

Conclusion. The derivatives are 12, 1, 0, and -11, respectively.

28.3 Exercise 28.3: Square roots and cube roots

What we must show. Derive the root differentiation formulas from limiting quotients and examine the cube root at zero.

(a) Rationalize the square-root quotient.

Fix $a > 0$. For $x \geq 0$ with $x \neq a$,

$$\frac{\sqrt{x} - \sqrt{a}}{x - a} = \frac{1}{\sqrt{x} + \sqrt{a}} \longrightarrow \frac{1}{2\sqrt{a}}.$$

Continuity of the square root justifies the last limit; its denominator tends to the positive number $2\sqrt{a}$. Thus $h'(a) = \frac{1}{2}a^{-1/2}$.

(b) Factor a difference of cubes.

Fix $a \neq 0$, and put $u = x^{1/3}$, $v = a^{1/3}$. Since $x - a = u^3 - v^3 = (u - v)(u^2 + uv + v^2)$,

$$\frac{x^{1/3} - a^{1/3}}{x - a} = \frac{1}{x^{2/3} + x^{1/3}a^{1/3} + a^{2/3}} \longrightarrow \frac{1}{3a^{2/3}}.$$

The real cube-root function is continuous, and $3a^{2/3} > 0$, even when $a < 0$. This proves $f'(a) = \frac{1}{3}a^{-2/3}$.

(c) The origin is different.

At zero the quotient is $x^{1/3}/x = 1/x^{2/3}$. It tends to $+\infty$ from both sides, not to a finite real number. A vertical tangent is not a derivative in the definition being used.

Conclusion. The formulas hold on the stated nonzero domains; the cube-root function is not differentiable at 0.

28.4 Exercise 28.4: A differentiable function with a discontinuous derivative

What we must show. Analyze $f(x) = x^2 \sin(1/x)$ off zero, with $f(0) = 0$.

(a) Differentiate only where the displayed formula is valid.

For $a \neq 0$, the product and chain rules give

$$f'(a) = 2a \sin(1/a) + a^2 \cos(1/a)(-a^{-2}) = 2a \sin(1/a) - \cos(1/a).$$

The reciprocal, sine, and polynomial factors are differentiable near such an a, so the rules apply.

(b) Return to the definition at zero.

For $x \neq 0$,

$$\frac{f(x) - f(0)}{x} = x \sin(1/x), \qquad |x \sin(1/x)| \leq |x| \longrightarrow 0.$$

Thus $f'(0) = 0$.

(c) Test continuity using a particular sequence.

Take $x_n = 1/(2\pi n) \to 0$. Then $\sin(1/x_n) = 0$ and $\cos(1/x_n) = 1$, so $f'(x_n) = -1$ for every n. These values do not tend to $f'(0) = 0$. Therefore f' is not continuous at zero.

Conclusion. The function is differentiable everywhere, but its derivative fails to be continuous at zero.

28.5 Exercise 28.5: Why an informal chain-rule cancellation can fail

What we must show. Explain the problem with a quotient involving $f(x) - f(0)$ for the function in Exercise 28.4 and $g(x) = x$.

(a) Verify the separate derivatives exist.

Exercise 28.4 proves that f is differentiable everywhere, including zero. The identity function g has derivative 1 everywhere.

(b) Locate infinitely many zeros.

For any nonzero integer n, let $x = 1/(\pi n)$. Then

$$f(x) = \frac{1}{\pi^2 n^2} \sin(\pi n) = 0 = f(0).$$

Such nonzero x occur arbitrarily close to zero.

(c) Inspect the proposed quotient before taking a limit.

At every one of these points,

$$\frac{g(f(x)) - g(f(0))}{f(x) - f(0)} = \frac{0}{0},$$

which is undefined. Thus it is not a function on a full deleted neighborhood of zero, as a naive factorization of the chain-rule difference quotient would require. This is the sense in which the displayed unrestricted limit is meaningless in the exercise. If one explicitly restricts its domain to points with $f(x) \neq f(0)$, the quotient equals 1 and has relative limit 1; that different statement does not make the missing values defined. A rigorous chain-rule proof fills in the outer difference quotient at the problematic value using $g'(f(0))$, or uses a differentiability remainder.

Conclusion. Differentiability of f does not guarantee $f(x) \neq f(0)$ for nearby $x \neq 0$. Division by this difference needs separate justification.

28.6 Exercise 28.6: Continuity without differentiability at an oscillating point

What we must show. Examine $f(x) = x \sin(1/x)$ for $x \neq 0$, with $f(0) = 0$.

(a) Continuity.

The estimate $|f(x) - f(0)| \leq |x|$ shows that $f(x) \to f(0)$ as $x \to 0$. For an explicit epsilon–delta proof, choose $\delta = \varepsilon$.

(b) Differentiability.

The difference quotient at zero is $\sin(1/x)$. Along

$$u_n = \frac{1}{\pi/2 + 2\pi n}, \qquad v_n = \frac{1}{3\pi/2 + 2\pi n},$$

both sequences tend to zero, but their quotient values are 1 and -1, respectively. A function with a limit must have the same limit along every approaching sequence. Hence the derivative does not exist.

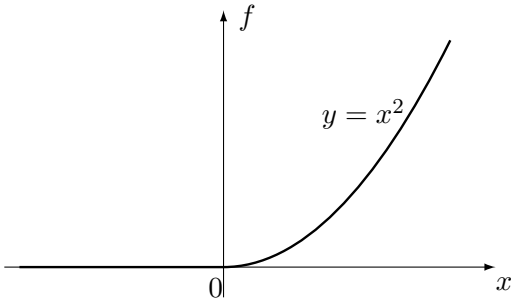
Conclusion. The factor x controls the function values, but dividing by x in the derivative removes precisely that control.

28.7 Exercise 28.7: Joining a horizontal half-line to a parabola

What we must show. For $f(x) = 0$ on $x < 0$ and $f(x) = x^2$ on $x \geq 0$, draw and analyze f and f'.

(a) Sketch the function.

The left branch is the negative half of the x-axis. The right branch is the right half of the parabola $y = x^2$, joined at $(0, 0)$.



(b) Differentiate at the join from the definition.

For $h \neq 0$,

$$\frac{f(h) - f(0)}{h} = \begin{cases} 0, & h < 0, \\ h, & h > 0. \end{cases}$$

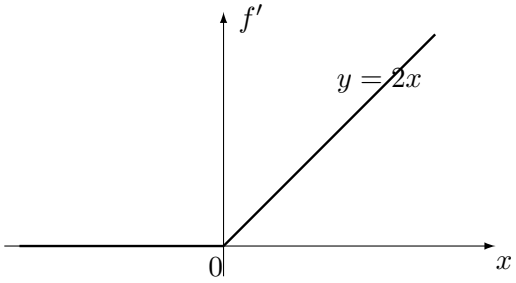
Both sides tend to zero, so $f'(0) = 0$.

(c) Calculate and sketch the derivative.

On the open half-lines, ordinary differentiation gives

$$f'(x) = \begin{cases} 0, & x \leq 0, \\ 2x, & x > 0. \end{cases}$$

Its graph is a horizontal left half-line joined to a straight right half-line of slope 2:



(d) Separate continuity from another derivative.

Both branches of f' approach zero at the origin, its actual value there. Thus f' is continuous on all of $\mathbb{R}$. Its difference quotient at zero, however, is 0 for $h < 0$ and 2 for $h > 0$. These do not agree. Consequently f' is differentiable on $\mathbb{R} \setminus \{0\}$ but not on all of $\mathbb{R}$.

Conclusion. f is continuously differentiable, but its second derivative does not exist at zero.

28.8 Exercise 28.8: A rational–irrational example

What we must show. Study $f(x) = x^2$ at rational x and $f(x) = 0$ at irrational x.

(a) Continuity at zero uses a common bound.

In both cases $0 \le f(x) \le x^2$, and $f(0) = 0$. Given $\varepsilon > 0$, choosing $\delta = \sqrt{\varepsilon}$ makes $|x| < \delta$ imply $|f(x) - f(0)| < \varepsilon$.

(b) At a nonzero point, compare dense kinds of inputs.

Fix $a \neq 0$. Density supplies rational $r_n \to a$ and irrational $t_n \to a$. Then

$$f(r_n) = r_n^2 \longrightarrow a^2, \qquad f(t_n) = 0 \longrightarrow 0.$$

Since $a^2 \neq 0$, these two limits disagree. Thus f has no limit at a and cannot be continuous there.

(c) Differentiate at zero without assuming a neighborhood formula.

For $h \neq 0$ the quotient is h if h is rational and 0 otherwise. In either case its absolute value is at most $|h|$, so its limit is zero. Hence $f'(0) = 0$. At every other point differentiability fails because continuity already fails.

Conclusion. The derivative exists exactly at 0, where it equals 0. It would be invalid to claim $f'(x) = 2x$ just because one branch has that formula.

28.9 Exercise 28.9: Separating an inner polynomial and an outer power

What we must show. Differentiate $h(x) = (x^4 + 13x)^7$ and identify the composition.

(a) Compute the derivative.

The answer is

$$h'(x) = 7(x^4 + 13x)^6(4x^3 + 13).$$

The factor $4x^3 + 13$ is essential: it accounts for the rate of change of the inner quantity.

(b) Justify each factor.

Set $f(x) = x^4 + 13x$ and $g(u) = u^7$. These are polynomials, so they are differentiable everywhere. Since $h = g \circ f$, the chain rule gives $h'(x) = g'(f(x))f'(x)$, with $g'(u) = 7u^6$ and $f'(x) = 4x^3 + 13$. Substitution gives part (a).

28.10 Exercise 28.10: A power of a trigonometric–exponential sum

What we must show. Repeat the preceding decomposition for $h(x) = (\cos x + e^x)^{12}$.

(a) Differentiate.

$$h'(x) = 12(\cos x + e^x)^{11}(-\sin x + e^x).$$

(b) Verify the chain rule applies.

Let $f(x) = \cos x + e^x$ and $g(u) = u^{12}$. Both are differentiable everywhere, $f'(x) = -\sin x + e^x$, and $g'(u) = 12u^{11}$. Thus $h = g \circ f$ and the displayed formula is exactly $g'(f(x))f'(x)$.

28.11 Exercise 28.11: A chain of three functions

What we must show. State and prove the derivative rule for $h \circ g \circ f$ at a under the given differentiability assumptions.

First combine the two inner functions.

Since f is differentiable at a and g at $f(a)$, the chain rule makes $g \circ f$ differentiable at a and gives

$$(g \circ f)'(a) = g'(f(a))f'(a).$$

Differentiability also gives the continuity needed to ensure the compositions are defined near the relevant points.

Then apply the rule to the remaining outer function.

The assumption about h is differentiability at $(g \circ f)(a)$. A second application yields

$$(h \circ g \circ f)'(a) = h'(g(f(a)))(g \circ f)'(a) = h'(g(f(a)))g'(f(a))f'(a).$$

Conclusion. Each derivative is evaluated at the input that its particular function actually receives.

28.12 Exercise 28.12: Three nested functions with an exponent

What we must show. Differentiate $H(x) = \cos(e^{x^5-3x})$; the entire expression $x^5 - 3x$ is in the exponent.

(a) Work from the outside inward.

Differentiating cosine introduces a minus sine; differentiating the exponential reproduces it; differentiating the polynomial gives $5x^4 - 3$. Therefore

$$H'(x) = -\sin(e^{x^5-3x})\, e^{x^5-3x}(5x^4 - 3).$$

(b) Make the three-function justification explicit.

Take $f(x) = x^5 - 3x$, $g(u) = e^u$, and $h(v) = \cos v$. They are differentiable on $\mathbb{R}$, so the hypotheses of Exercise 28.11 hold at every x. Substituting f', g', and h' into its formula proves the calculation.

28.13 Exercise 28.13: Ensuring a composition is defined nearby

What we must show. Find an open interval around a on which $g \circ f$ is defined.

Choose a target neighborhood that stays in the domain of g.

Because that domain contains an open interval around $f(a)$, there is $\eta > 0$ such that $(f(a)-\eta, f(a)+\eta)$ lies in the domain of g.

Use continuity of f to stay inside that neighborhood.

Continuity at a supplies $\delta_1 > 0$ such that $|x-a| < \delta_1$ implies $|f(x)-f(a)| < \eta$, for x in the domain of f. Its domain also contains $(a-\delta_2, a+\delta_2)$ for some $\delta_2 > 0$. Let $\delta = \min(\delta_1, \delta_2)$. Every $x \in (a-\delta, a+\delta)$ belongs to the domain of f and has $f(x)$ in the domain of g. Thus $g(f(x))$ is defined throughout that open interval.

Conclusion. Continuity of g is unnecessary for this domain assertion; only its available domain matters.

28.14 Exercise 28.14: Increment quotients and symmetric quotients

What we must show. Derive both requested limits from differentiability at a.

(a) Replace the approaching point by an increment.

Set $x = a + h$. Then $x \to a$ if and only if $h \to 0$, and $x \neq a$ if and only if $h \neq 0$. Consequently the definition immediately gives

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = f'(a).$$

(b) Split the symmetric difference into two ordinary ones.

For nonzero h small enough that both arguments are in the domain,

$$\frac{f(a+h)-f(a-h)}{2h} = \frac{1}{2}\left(\frac{f(a+h)-f(a)}{h} + \frac{f(a-h)-f(a)}{-h}\right).$$

Each quotient in parentheses tends to $f'(a)$ by part (a), the second with increment $-h$. The limit is therefore $\frac{1}{2}(f'(a)+f'(a)) = f'(a)$.

Conclusion. Differentiability implies the symmetric limit formula. The converse need not hold: $|x|$ at zero has symmetric quotient 0 but no derivative.

28.15 Exercise 28.15: Leibniz's higher product rule

What we must show. Prove the binomial-coefficient formula for the nth derivative of a product.

Plan. Induction extends the usual product rule. Each differentiation can land on either factor; Pascal's identity combines the two contributions to an interior term.

Start the induction.

For $n = 1$ the formula is $(fg)' = fg' + f'g$, precisely the product rule. We use $f^{(0)} = f$ and $g^{(0)} = g$.

Differentiate the induction formula.

Assume the identity for order n, and suppose the derivatives needed for order $n+1$ exist. On the neighborhood where the lower derivatives are defined, differentiate each term:

$$(fg)^{(n+1)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k+1)} g^{(n-k)} + \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n+1-k)}.$$

Reindex the first sum with $j = k + 1$. Its end term at $j = n + 1$ is $f^{(n+1)}g$, and the second sum's end term at $j = 0$ is $fg^{(n+1)}$. For $1 \le j \le n$, the coefficient of $f^{(j)}g^{(n+1-j)}$ is

$$\binom{n}{j-1} + \binom{n}{j} = \binom{n+1}{j}.$$

The two end coefficients are also the corresponding binomial coefficients, both 1. Thus

$$(fg)^{(n+1)}(a) = \sum_{j=0}^{n+1} \binom{n+1}{j} f^{(j)}(a)g^{(n+1-j)}(a).$$

This closes the induction. If only the final derivative is assumed at a, the same argument is applied to the lower-order identity near a and differentiated at a; it does not assume the highest derivatives exist throughout a larger interval.

Conclusion. For every permitted order, $(fg)^{(n)}(a) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(a)g^{(n-k)}(a)$.

28.16 Exercise 28.16: Differentiability as a linear approximation with vanishing error

What we must show. Prove the equivalence in the exercise, interpreting the proposed coefficient as a fixed real number A until differentiability has been established.

From a derivative to an error function.

If $f'(a)$ exists, let $A = f'(a)$ and define

$$\epsilon(x) = A - \frac{f(x) - f(a)}{x - a} \quad (x \neq a), \qquad \epsilon(a) = 0.$$

By the derivative definition, $\epsilon(x) \to 0$ as $x \to a$. Multiplication by $x - a$ gives

$$f(x) - f(a) = (x - a)[A - \epsilon(x)]$$

for $x \neq a$; at a both sides are zero, so the identity holds on all of I.

From the error representation to a derivative.

Conversely, suppose such a real A and such an error function are given. For $x \neq a$, divide the identity by $x - a$. The quotient becomes $A - \epsilon(x)$ and tends to A. Thus $f'(a)$ exists and equals A, making the notation $f'(a)$ in the exercise legitimate.

Conclusion. The change in f equals its linear part plus an error whose ratio to $x - a$ tends to zero. No continuity of the error away from a is required.

29 The Mean Value Theorem

The Mean Value Theorem (MVT) connects an average slope to an actual derivative. If f is continuous on $[a, b]$ and differentiable on (a, b), with $a < b$, there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

The theorem asserts existence, not a prescribed choice of c. Both hypotheses matter. Ross uses "increasing" to mean nondecreasing; "strictly increasing" requires strict inequalities. A derivative sign on an entire interval controls monotonicity there; a sign at one point does not have the same force.

29.1 Exercise 29.1: Checking the MVT conclusion in six examples

What we must show. Compute the secant slope and determine whether an interior derivative equals it.

(a) The square on $[-1, 2]$.

The slope is $(4 - 1)/3 = 1$. Since $f'(x) = 2x$, take $c = 1/2 \in (-1, 2)$. Polynomial continuity and differentiability verify all MVT hypotheses.

(b) Sine on $[0, \pi]$.

The secant slope is 0, and $f'(\pi/2) = \cos(\pi/2) = 0$. Sine satisfies all hypotheses.

(c) Absolute value on $[-1, 2]$.

The secant slope is $(2 - 1)/3 = 1/3$. At nonzero interior points the derivative is either -1 or 1; at 0 it is undefined. Thus no derivative equals $1/3$. The failed hypothesis is differentiability throughout the open interval, although continuity holds.

(d) The reciprocal on $[-1, 1]$.

The endpoint expression gives $(1 - (-1))/2 = 1$. Wherever it exists, the derivative is $-1/x^2 < 0$, so it cannot equal 1. The function is not even defined at 0; in particular it is neither continuous on the whole closed interval nor differentiable on the whole open interval.

(e) The reciprocal on $[1, 3]$.

Now both hypotheses hold. The slope is $(1/3 - 1)/2 = -1/3$, and $-1/c^2 = -1/3$ is satisfied by $c = \sqrt{3} \in (1, 3)$.

(f) The sign function on $[-2, 2]$.

The secant slope is $(1 - (-1))/4 = 1/2$. At every nonzero point its derivative is 0; at 0 it has a jump and no derivative. Thus the conclusion fails. Continuity on the closed interval and differentiability throughout the open interval both fail.

Conclusion. The conclusion holds in (a), (b), and (e), at the exhibited points; it fails in (c), (d), and (f).

29.2 Exercise 29.2: Cosine is Lipschitz with constant one

What we must show. Prove $|\cos x - \cos y| \le |x - y|$ for all real x, y.

Handle equal points and order unequal ones.

If $x = y$, both sides vanish. Otherwise interchange their names if needed so $x < y$. Cosine is continuous on $[x, y]$ and differentiable on (x, y). The MVT gives $c \in (x, y)$ with

$$\cos y - \cos x = -\sin c\,(y - x).$$

Taking absolute values and using $|\sin c| \le 1$ proves the inequality. The absolute-value form is unchanged when the original points are interchanged.

29.3 Exercise 29.3: Two guaranteed derivative values

What we must show. Use $f(0) = 0$, $f(1) = f(2) = 1$ to locate derivative values $1/2$ and $1/7$.

(a) Use the endpoints 0 and 2.

Differentiability on $\mathbb{R}$ implies continuity there, so the MVT applies on $[0, 2]$. It supplies $c \in (0, 2)$ with

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{1}{2}.$$

(b) First find slopes above and below 1/7.

The MVT on $[0, 1]$ gives $u \in (0, 1)$ with $f'(u) = 1$. On $[1, 2]$ it gives $v \in (1, 2)$ with $f'(v) = 0$. Derivatives have the intermediate value property (Darboux's theorem, Theorem 29.8), even when they are not continuous. Therefore some $w \in (u, v) \subset (0, 2)$ satisfies $f'(w) = 1/7$.

An elementary way to implement the same argument.

Let $H(x) = f(x) - x/7$ on $[u, v]$. Then $H'(u) = 6/7 > 0$ and $H'(v) = -1/7 < 0$. By the derivative definition, H takes values larger than $H(u)$ just to the right of u, and values larger than $H(v)$ just to the left of v. Its maximum on the compact interval therefore cannot occur at either endpoint. At an interior maximum w, Fermat's theorem gives $H'(w) = 0$, or $f'(w) = 1/7$. This explains why continuity of f' is not needed.

29.4 Exercise 29.4: Introducing an exponential integrating factor

What we must show. Find $c \in (a, b)$ with $f'(c) + f(c)g'(c) = 0$ when $f(a) = f(b) = 0$.

Plan. The derivative of fe^g contains exactly the desired expression, multiplied by a positive factor.

Verify Rolle's theorem for the auxiliary function.

Define $H(x) = f(x)e^{g(x)}$. The differentiability assumptions on the open interval I ensure that H is continuous on $[a, b] \subset I$ and differentiable on (a, b). Also $H(a) = H(b) = 0$. Rolle's theorem gives $c \in (a, b)$ with $H'(c) = 0$.

Differentiate and divide by a nonzero factor.

By the product and chain rules,

$$0 = H'(c) = e^{g(c)}[f'(c) + f(c)g'(c)].$$

Because $e^{g(c)} > 0$, division yields the required equation.

29.5 Exercise 29.5: A quadratic bound forces constancy

What we must show. Show a function satisfying $|f(x) - f(y)| \le (x-y)^2$ for all real x, y is constant.

Prove differentiability rather than assume it.

Fix $a \in \mathbb{R}$. For $h \neq 0$, the assumed estimate gives

$$\left|\frac{f(a+h) - f(a)}{h}\right| \le \frac{h^2}{|h|} = |h| \longrightarrow 0.$$

Thus $f'(a)$ exists and equals 0. Since a was arbitrary, $f' = 0$ everywhere.

Use the MVT between any two points.

For $x < y$, the now-established differentiability gives the MVT hypotheses on $[x, y]$. Hence $f(y) - f(x) = f'(c)(y - x) = 0$ for some c between them. Every pair has equal values, so f is constant.

29.6 Exercise 29.6: The straight line in the MVT proof

What we must show. Write the line through $(a, f(a))$ and $(b, f(b))$ explicitly.

Use point–slope form.

The required function is

$$L(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Indeed $L(a) = f(a)$ and $L(b) = f(a) + f(b) - f(a) = f(b)$. Its derivative is the constant secant slope $(f(b) - f(a))/(b - a)$. Subtracting this line from f gives a function with equal endpoint values, which is precisely why Rolle's theorem can be used in the MVT proof.

29.7 Exercise 29.7: Vanishing higher derivatives imply polynomial form

What we must show. Determine f when its second or third derivative vanishes on an interval.

(a) If $f'' = 0$, first make f' constant.

The zero-derivative corollary to the MVT, applied to f', gives a constant A with $f'(x) = A$ on I. Now $[f(x) - Ax]' = 0$ on I, so the same corollary gives another constant B with $f(x) - Ax = B$. Thus $f(x) = Ax + B$.

(b) If $f''' = 0$, repeat one level higher.

Apply the corollary to f'' to obtain $f''(x) = C$. Then

$$[f'(x) - Cx]' = 0,$$

so $f'(x) = Cx + D$. Finally,

$$\left[f(x) - \frac{C}{2}x^2 - Dx\right]' = 0,$$

which gives $f(x) = \frac{C}{2}x^2 + Dx + E$. Conversely, every polynomial of this form has third derivative zero, so these are exactly the possibilities.

Conclusion. The interval assumption allows the MVT to connect every pair of points. On disconnected domains the constants could differ between components.

29.8 Exercise 29.8: Derivative signs and the three remaining monotonicity rules

What we must show. Prove Corollary 29.7(ii)–(iv).

Use one MVT equation for all three cases.

Choose arbitrary $x_1 < x_2$ in the interval. The MVT gives $c \in (x_1, x_2)$ with

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

The factor $x_2 - x_1$ is strictly positive, so it preserves the sign of $f'(c)$.

(ii) Strictly negative derivative.

If $f' < 0$ everywhere, the right side is negative. Thus $f(x_2) < f(x_1)$ for every ordered pair, which is strict decrease.

(iii) Nonnegative derivative.

If $f' \geq 0$ everywhere, then $f(x_2) - f(x_1) \geq 0$, so $f(x_1) \leq f(x_2)$. This is increase in Ross's nonstrict sense.

(iv) Nonpositive derivative.

If $f' \leq 0$ everywhere, the equation gives $f(x_2) \leq f(x_1)$, which is decrease. Equality is permitted in the last two cases.

29.9 Exercise 29.9: The inequality $ex \leq e^x$

What we must show. Show the product $e \cdot x$ never exceeds e^x.

Subtract and locate the minimum.

Let $F(x) = e^x - ex$. Then $F'(x) = e^x - e$. Because the exponential is strictly increasing, this derivative is negative for $x < 1$, zero at 1, and positive for $x > 1$. The monotonicity corollaries imply F decreases toward 1 and increases afterward. Therefore its global minimum is

$$F(1) = e - e = 0.$$

Hence $e^x - ex \geq 0$ for every real x.

Conclusion. The inequality holds for every real x, with equality exactly when $x = 1$. Here ex denotes multiplication, not a second copy of e^x.

29.10 Exercise 29.10: A positive derivative at one point does not imply local increase

What we must show. Study $f(x) = x^2 \sin(1/x) + x/2$ for $x \neq 0$, with $f(0) = 0$.

(a) Differentiate at zero.

The difference quotient is $x \sin(1/x) + 1/2$, whose first term tends to zero by $|x \sin(1/x)| \leq |x|$. Thus $f'(0) = 1/2 > 0$.

(b) Find negative derivatives arbitrarily close to zero.

For $x \neq 0$, Exercise 28.4 and the sum rule give

$$f'(x) = 2x \sin(1/x) - \cos(1/x) + \frac{1}{2}.$$

At $x_n = 1/(2\pi n)$ this equals $-1/2$. Every open interval containing zero contains such an x_n as an interior point. An increasing function that is differentiable at an interior point has nonnegative derivative there: its difference quotients are nonnegative for increments of either sign. Thus the negative derivative at x_n rules out increase on that interval.

(c) Compare the hypotheses carefully.

Corollary 29.7(i) requires $f'(x) > 0$ at *every* point of the interval. This example gives positivity only at zero and negative values arbitrarily nearby, so there is no contradiction.

29.11 Exercise 29.11: Sine lies below its argument on the positive half-line

What we must show. Prove $\sin x \leq x$ for $x \geq 0$.

Apply a derivative sign to the difference.

Let $F(x) = x - \sin x$. Then $F'(x) = 1 - \cos x \geq 0$ on $\mathbb{R}$, so F is increasing. For $x \geq 0$ this implies $F(x) \geq F(0) = 0$, which rearranges to the required inequality. At $x = 0$ equality is immediate.

29.12 Exercise 29.12: Three related trigonometric inequalities

What we must show. Establish the tangent inequality, monotonicity of $x/\sin x$, and the sine lower bound on a quarter period.

(a) Compare tangent with the identity.

Let $F(x) = \tan x - x$ on $[0, \pi/2)$. It is continuous at zero, $F(0) = 0$, and for $0 < x < \pi/2$,

$$F'(x) = \sec^2 x - 1 = \tan^2 x > 0.$$

Applying the MVT on $[0, x]$ gives $F(x) > F(0) = 0$. Thus $x < \tan x$.

(b) Differentiate the quotient with a positive denominator.

On $(0, \pi/2)$,

$$\left(\frac{x}{\sin x}\right)' = \frac{\sin x - x \cos x}{\sin^2 x}.$$

Both $\cos x$ and $\sin^2 x$ are positive. Part (a), multiplied by $\cos x > 0$, gives $x \cos x < \sin x$. Therefore the numerator is positive, and the quotient is strictly increasing.

(c) Compare with the endpoint value.

The quotient extends continuously to $x = \pi/2$ with value $\pi/2$. Its increase on the interior yields $x/\sin x \leq \pi/2$ for $0 < x \leq \pi/2$, for instance by taking an increasing sequence of comparison points tending to the endpoint. Multiplying by the positive $\sin x$ gives $x \leq (\pi/2)\sin x$. At $x = 0$ both sides vanish.

Conclusion. The last inequality is equivalently $\sin x \geq 2x/\pi$ on $[0, \pi/2]$.

29.13 Exercise 29.13: Comparing functions by comparing derivatives

What we must show. Use equal initial values and $f' \leq g'$ to show $f(x) \leq g(x)$ for $x \geq 0$.

Build a single function with a known sign of derivative.

Set $H = g - f$. Then $H(0) = 0$ and $H' = g' - f' \geq 0$. The monotonicity corollary makes H increasing. Hence for $x \geq 0$, $H(x) \geq H(0) = 0$, or $f(x) \leq g(x)$.

Conclusion. Equal initial values anchor the comparison. Without them, derivative inequalities alone do not fix which graph starts higher.

29.14 Exercise 29.14: Bounding a function between two straight lines

What we must show. Given $f(0) = 0$ and $1 \leq f' \leq 2$, prove $x \leq f(x) \leq 2x$ for $x \geq 0$.

Use the MVT on the interval from zero to the chosen point.

At $x = 0$ the assertion is immediate. For $x > 0$, differentiability ensures the hypotheses on $[0, x]$, so for some $c \in (0, x)$,

$$f(x) - f(0) = f'(c)x.$$

Since $x > 0$, multiplying $1 \leq f'(c) \leq 2$ by x preserves order. Substituting $f(0) = 0$ gives $x \leq f(x) \leq 2x$.

29.15 Exercise 29.15: The power rule for rational exponents

What we must show. Prove the derivative formula for $h(x) = x^{m/n}$ at every permitted nonzero real input, including negative inputs when n is odd.

Differentiate the real root by the inverse-function theorem.

Let $u(x) = x^{1/n}$ on its real domain. Fix a permitted $a \neq 0$ and write $t = u(a)$, so $t^n = a$ and $t \neq 0$. The function $v(t) = t^n$, restricted to the positive half-line if n is even and considered on $\mathbb{R}$ if n is odd, is one-to-one and continuous. Near this nonzero t it is differentiable with $v'(t) = nt^{n-1} \neq 0$. The inverse derivative theorem gives

$$u'(a) = \frac{1}{nt^{n-1}}.$$

Compose with an integer power.

The function $h = u^m$ is differentiable near a; negative m is allowed because u is nonzero there. The integer power rule and chain rule give

$$h'(a) = mt^{m-1}\frac{1}{nt^{n-1}} = \frac{m}{n}t^{m-n} = \frac{m}{n}a^{m/n-1}.$$

The final equality uses the real-root definition of the rational powers. It remains valid for negative a when n is odd, without invoking an undefined logarithm of a negative number.

Conclusion. $h'(a) = ra^{r-1}$ everywhere requested. The exclusion of zero avoids both inverse-derivative failure and negative-power domain issues.

29.16 Exercise 29.16: The derivative of arctangent

What we must show. Apply the inverse derivative theorem to tangent on $(-\pi/2, \pi/2)$.

Check invertibility and derivative hypotheses.

On this interval, $f(x) = \tan x$ is continuous and strictly increasing, has range $\mathbb{R}$, and has derivative $\sec^2 x > 0$. It therefore has the inverse $g = \arctan : \mathbb{R} \to (-\pi/2, \pi/2)$, and Theorem 29.9 applies at every inverse pair $y = \tan x$.

Express the derivative using the new input.

The theorem gives $g'(y) = 1/\sec^2 x$. Since $\sec^2 x = 1 + \tan^2 x = 1 + y^2$, this becomes

$$\frac{d}{dy}\arctan y = \frac{1}{1+y^2} \qquad (y \in \mathbb{R}).$$

29.17 Exercise 29.17: Matching values and slopes at a join

What we must show. Prove the criterion for differentiability of h, formed from f to the left of a and g at and to its right.

Necessity of matching values.

Suppose h is differentiable at a. It is then continuous there. By continuity of f,

$$f(a) = \lim_{x\to a^-} f(x) = \lim_{x\to a^-} h(x) = h(a) = g(a).$$

Thus the two branches meet at the same height.

Necessity of matching slopes.

Using $h(a) = f(a) = g(a)$, the left difference quotient for h has limit $f'(a)$, while the right one has limit $g'(a)$. Both must equal $h'(a)$, so $f'(a) = g'(a)$.

Sufficiency.

Conversely, suppose both equalities hold. For $x < a$ the difference quotient of h at a is $(f(x) - f(a))/(x - a)$; for $x > a$ it is $(g(x) - g(a))/(x - a)$. Their limits are the equal numbers $f'(a)$ and $g'(a)$. The two-sided limit therefore exists, and $h'(a)$ is their common value.

Conclusion. Matching heights prevents a jump; matching slopes prevents a corner. Both conditions are necessary and together are sufficient.

29.18 Exercise 29.18: Iterating a contraction on the real line

What we must show. Prove convergence of $s_{n+1} = f(s_n)$ and obtain a fixed point when $a = \sup_{x\in\mathbb{R}} |f'(x)| < 1$.

(a), first stage: convert the derivative bound to a distance bound.

Since $|f'| \geq 0$, we have $0 \leq a < 1$. The MVT gives, for any $u, v \in \mathbb{R}$,

$$|f(u) - f(v)| \leq a|u - v|.$$

For equal inputs the inequality is automatic. Applying it to successive iterates gives

$$|s_{n+1} - s_n| \leq a|s_n - s_{n-1}| \leq \cdots \leq a^n|s_1 - s_0|.$$

(a), second stage: sum the small successive gaps.

Assume first $0 < a < 1$. For integers $m > n$, the triangle inequality and a geometric series give

$$|s_m - s_n| \leq \sum_{k=n}^{m-1} |s_{k+1} - s_k| \leq |s_1 - s_0| \sum_{k=n}^{m-1} a^k \leq \frac{|s_1 - s_0|a^n}{1 - a}.$$

This bound tends to zero independently of $m > n$. For any $\varepsilon > 0$, choose N to make the last expression less than ε for $n \geq N$; this is exactly the Cauchy criterion. Completeness of $\mathbb{R}$ gives a limit s. If $a = 0$, the distance bound says f is constant, so all iterates from s_1 onward are equal and convergence is immediate.

(b) Pass to the limit in the iteration.

Differentiability implies continuity, and therefore

$$f(s) = f\left(\lim_n s_n\right) = \lim_n f(s_n) = \lim_n s_{n+1} = s.$$

Thus s is a fixed point. In fact it is unique: if $t = f(t)$, then $|s - t| = |f(s) - f(t)| \leq a|s - t|$; since $1 - a > 0$, this forces $s = t$.

Conclusion. Every starting value produces a convergent sequence with the same fixed-point limit. The geometric bound explicitly explains why the iteration settles down.

30 L'Hospital's rule

L'Hospital's rule compares a quotient with a quotient of derivatives; it does not say that every limit may be evaluated by differentiating. In the 0/0 case, both numerator and denominator must tend to zero. Ross also permits the case in which the absolute value of the denominator tends to infinity, without requiring a numerator limit. In either case the functions must be differentiable on the relevant deleted neighborhood, the derivative of the denominator must be nonzero there, and the derivative quotient must have a limit. That limit may be finite or infinite.

For positive quantities with variable exponents, take a logarithm first. If $A(x) > 0$, then $A(x)^{B(x)} = \exp(B(x) \log A(x))$; after finding the exponent's limit, use continuity of the exponential.

30.1 Exercise 30.1: Four introductory limits

What we must show. Evaluate each limit and justify the method, using the actual numerator and denominator in the printed exercise.

(a) The quotient $(e^{2x} - \cos x)/x$ at zero.

Both numerator and denominator tend to zero, they are differentiable, and the denominator derivative is $1 \neq 0$. L'Hospital's rule therefore gives

$$\lim_{x \to 0} \frac{e^{2x} - \cos x}{x} = \lim_{x \to 0} (2e^{2x} + \sin x) = 2.$$

(b) The quotient $(1 - \cos x)/x^2$ at zero.

The original form is 0/0, and the derivative of x^2 is nonzero for $x \neq 0$. The derivative quotient has the known limit

$$\lim_{x \to 0} \frac{\sin x}{2x} = \frac{1}{2}.$$

L'Hospital's rule thus gives 1/2 for the original limit. The last step uses the basic sine limit, not another unjustified cancellation at zero.

(c) The quotient x^3/e^{2x} at positive infinity.

At every differentiation stage the exponential denominator tends to infinity and has nonzero derivative. Successive derivative quotients reduce the polynomial degree:

$$\frac{x^3}{e^{2x}}, \quad \frac{3x^2}{2e^{2x}}, \quad \frac{6x}{4e^{2x}}, \quad \frac{6}{8e^{2x}}.$$

The final quotient tends to zero. Applying the rule backward through these three stages proves that the original limit is 0.

(d) A difference of square roots at zero.

For $0 < |x| < 1$, multiply by the conjugate:

$$\frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} = \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \longrightarrow 1.$$

This direct algebraic method is shorter than L'Hospital's rule and also keeps the real domain explicit.

Conclusion. The four limits are 2, 1/2, 0, and 1.

30.2 Exercise 30.2: Cancellation, repeated differentiation, and a variable power

What we must show. Evaluate the four limits, checking why each reduction is valid.

(a) The quotient $x^3/(\sin x - x)$.

The numerator and denominator and their first two derivative pairs tend to zero. Differentiating each pair produces

$$\frac{x^3}{\sin x - x}, \quad \frac{3x^2}{\cos x - 1}, \quad \frac{6x}{-\sin x}, \quad \frac{6}{-\cos x}.$$

On a sufficiently small deleted neighborhood of zero, each denominator derivative used in passing to the next pair is nonzero: these are $\cos x - 1$, $-\sin x$, and $-\cos x$. The final quotient tends to -6, so three valid applications of L'Hospital's rule yield -6.

(b) The quotient $(\tan x - x)/x^3$.

This is $0/0$ near zero, where tangent is defined. The denominator derivative $3x^2$ is nonzero for $x \neq 0$. The derivative quotient is

$$\frac{\sec^2 x - 1}{3x^2} = \frac{1}{3}\left(\frac{\tan x}{x}\right)^2 \longrightarrow \frac{1}{3},$$

because $\tan x/x = (\sin x/x)/\cos x \to 1$. Thus the requested limit is $1/3$.

(c) The difference $1/\sin x - 1/x$.

Combine the fractions before applying any limit rule:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

The first derivative quotient is $(1 - \cos x)/(\sin x + x\cos x)$, again of form $0/0$. Its derivative quotient is

$$\frac{\sin x}{2\cos x - x \sin x} \longrightarrow 0.$$

The final denominator tends to 2 and hence is nonzero near zero. The intermediate denominator $\sin x + x\cos x$ is nonzero for sufficiently small nonzero x, because its ratio to x tends to 2. Thus both uses of L'Hospital's rule are legitimate and the original limit is 0.

(d) The expression $(\cos x)^{1/x^2}$.

For $|x| < \pi/2$, the base is positive, so its logarithm is $\log(\cos x)/x^2$. This is $0/0$, and its derivative quotient is

$$\frac{-\tan x}{2x} \longrightarrow -\frac{1}{2}.$$

L'Hospital's rule gives the logarithmic limit $-1/2$. Continuity of the exponential now gives the original limit $e^{-1/2}$.

Conclusion. The answers are -6, $1/3$, 0, and $e^{-1/2}$.

30.3 Exercise 30.3: Recognizing when differentiation is unnecessary

What we must show. Evaluate the four limits, including a correction to a printed selected answer.

(a) The quotient $(x - \sin x)/x$ at positive infinity.

For the expression printed in the exercise,

$$\frac{x - \sin x}{x} = 1 - \frac{\sin x}{x}, \qquad \left|\frac{\sin x}{x}\right| \le \frac{1}{x} \longrightarrow 0.$$

The limit is therefore 1. The selected-answer entry 0 in the linked edition is inconsistent with this printed expression. Differentiating would give $1 - \cos x$, which has no limit; this simply means L'Hospital's rule is not useful here. Its failure to supply a derivative-quotient limit says nothing against the elementary estimate above.

(b) The power $x^{\sin(1/x)}$ at positive infinity.

Its logarithm is

$$\sin(1/x)\log x = \frac{\sin(1/x)}{1/x}\,\frac{\log x}{x}.$$

The first factor tends to 1. The second tends to 0 by L'Hospital's rule, since $(\log x)'/x' = 1/x \to 0$ and $x \to \infty$. Thus the logarithm tends to zero and the power tends to $e^0 = 1$.

(c) The quotient $(1 + \cos x)/(e^x - 1)$ as $x \to 0^+$.

The numerator tends to 2, while the denominator tends to zero through positive values. Consequently the limit is $+\infty$. Explicitly, near zero the numerator is greater than 1. For any prescribed $M > 0$, continuity of e^x makes $0 < e^x - 1 < 1/M$ for sufficiently small $x > 0$, giving the quotient greater than M. This is not a 0/0 situation.

(d) The quotient $(1 - \cos 2x - 2x^2)/x^4$ at zero.

For transparency, list all derivatives through the first nonzero limiting values:

order	numerator derivative	denominator derivative
0	$1 - \cos 2x - 2x^2$	x^4
1	$2\sin 2x - 4x$	$4x^3$
2	$4\cos 2x - 4$	$12x^2$
3	$-8\sin 2x$	$24x$
4	$-16\cos 2x$	24

At orders 0 through 3 both entries tend to zero. Every denominator derivative required by the rule is nonzero for nonzero x near zero. The last ratio tends to $-16/24 = -2/3$. Four applications, read backward, prove that this is the requested limit.

Conclusion. For the printed questions, the limits are 1, 1, $+\infty$, and $-2/3$.

30.4 Exercise 30.4: Changing a right-hand zero limit into an infinite limit

What we must show. For $g(y) = f(1/y)$, prove equivalence of the two limits, including infinite limits.

Plan. The substitution is reversible: $y \to +\infty$ corresponds exactly to $x = 1/y \to 0^+$.

From a right-hand limit of f to a limit of g.

Suppose first $f(x) \to L \in \mathbb{R}$ as $x \to 0^+$. Given $\varepsilon > 0$, choose $\delta > 0$, with $\delta < a$ if a is finite, such that $0 < x < \delta$ implies $|f(x) - L| < \varepsilon$. If $y > 1/\delta$, then $0 < 1/y < \delta$, so $|g(y) - L| < \varepsilon$. This proves $g(y) \to L$ as $y \to \infty$.

Reverse the argument.

Suppose $g(y) \to L \in \mathbb{R}$. Choose $Y > 0$ inside its domain such that $y > Y$ implies $|g(y) - L| < \varepsilon$. Let $\delta > 0$ be at most $1/Y$ and small enough that $(0, \delta) \subset (0, a)$. Then $0 < x < \delta$ implies $1/x > Y$, and

$$|f(x) - L| = |g(1/x) - L| < \varepsilon.$$

Thus $f(x) \to L$ as $x \to 0^+$.

Use the same substitution for infinite limits.

For a limit $+\infty$, replace $|f(x) - L| < \varepsilon$ and $|g(y) - L| < \varepsilon$ by $f(x) > M$ and $g(y) > M$, respectively, for arbitrary real thresholds M. The same reciprocal choices of Y and δ give both implications. For $-\infty$, use the inequalities $f(x) < M$ and $g(y) < M$. No other part of the proof changes.

Conclusion. Either limit exists exactly when the other does, and their finite or infinite values are equal. The convention $a^{-1} = 0$ when $a = \infty$ gives precisely the correct transformed domain.

30.5 Exercise 30.5: Three exponential limits

What we must show. Evaluate the limits by taking logarithms on domains where the bases are positive.

(a) The expression $(1 + 2x)^{1/x}$ as $x \to 0$.

For $|x| < 1/2$, the base is positive. Its logarithm is $\log(1 + 2x)/x$, of form $0/0$. The denominator derivative is 1, and the numerator derivative is $2/(1 + 2x) \to 2$. Thus the logarithm tends to 2, and the original expression tends to e^2.

(b) The expression $(1 + 2/y)^y$ as $y \to \infty$.

Set $x = 1/y \to 0^+$. The expression becomes $(1 + 2x)^{1/x}$, whose right-hand limit is e^2 by part (a). Exercise 30.4 rigorously justifies the reciprocal change of variable.

(c) The expression $(e^x + x)^{1/x}$ as $x \to \infty$.

The base is positive for $x > 0$. Factor out e^x inside its logarithm:

$$\frac{\log(e^x + x)}{x} = \frac{x + \log(1 + xe^{-x})}{x} = 1 + \frac{\log(1 + xe^{-x})}{x}.$$

L'Hospital's rule gives $x/e^x \to 0$, so $\log(1 + xe^{-x}) \to 0$. Dividing this term by $x \to \infty$ still gives zero; for example its absolute value is eventually at most $1/x$. The logarithmic limit is 1, and the original limit is e.

Conclusion. The three limits are e^2, e^2, and e.

30.6 Exercise 30.6: Recovering limits of a function and its derivative from their sum

What we must show. Given $f(x) + f'(x) \to L \in \mathbb{R}$ at positive infinity, prove $f(x) \to L$ and $f'(x) \to 0$.

Plan. Multiplication by e^x makes the sum appear as a product derivative.

Apply the denominator-infinity version of the rule.

Write

$$f(x) = \frac{f(x)e^x}{e^x}.$$

Both numerator and denominator are differentiable on (c, ∞), the denominator tends to $+\infty$, and its derivative e^x never vanishes. The derivative quotient is

$$\frac{(f(x)e^x)'}{(e^x)'} = \frac{e^x(f'(x) + f(x))}{e^x} = f'(x) + f(x) \longrightarrow L.$$

Theorem 30.2 therefore gives $f(x) \to L$. It is unnecessary to know whether the numerator itself has a limit: Ross's denominator-infinity formulation does not require that.

Subtract two now-known finite limits.

Since $f'(x) = [f(x) + f'(x)] - f(x)$, ordinary limit arithmetic gives $f'(x) \to L - L = 0$.

Conclusion. Both conclusions follow without assuming a limit of f or f' at the outset.

30.7 Exercise 30.7: Why zeros of the denominator derivative cannot be canceled away

What we must show. Analyze $f(x) = x + \cos x \sin x$ and $g(x) = e^{\sin x} f(x)$ and explain the apparent L'Hospital paradox.

(a) Both functions tend to positive infinity.

Since $|\cos x \sin x| \le 1$, we have $f(x) \ge x - 1 \to \infty$. Also $e^{\sin x} \ge e^{-1} > 0$, so for $x > 1$,

$$g(x) \ge e^{-1}(x - 1) \longrightarrow \infty.$$

(b) Differentiate and factor.

Using $1 = \sin^2 x + \cos^2 x$,

$$f'(x) = 1 - \sin^2 x + \cos^2 x = 2\cos^2 x.$$

The product and chain rules give

$$\begin{aligned} g'(x) &= e^{\sin x} \cos x \, f(x) + e^{\sin x} \, 2\cos^2 x \\ &= e^{\sin x} \cos x [2\cos x + f(x)]. \end{aligned}$$

(c) Cancel only at points where cancellation is permitted.

If $x > 3$, then $2\cos x + f(x) \geq -2 + x - 1 = x - 3 > 0$. When also $\cos x \neq 0$, the denominator derivative is nonzero and

$$\frac{f'(x)}{g'(x)} = \frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}.$$

If $\cos x = 0$, however, both derivatives vanish and the left side is undefined. The algebra does not define this original derivative quotient at those points.

(d) Distinguish a continuous extension from the actual quotient.

The expression on the right in part (c) is defined for every $x > 3$, and satisfies

$$\left|\frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}\right| \leq \frac{2e}{x-3} \longrightarrow 0.$$

Nevertheless, $g'(\pi/2 + k\pi) = 0$ for arbitrarily large integers k. Thus there is no tail on which g' is everywhere nonzero, as required by L'Hospital's rule. For the original functions, $f(x) > 0$ on $x > 1$, and cancellation of the nonzero $f(x)$ gives

$$\frac{f(x)}{g(x)} = e^{-\sin x}.$$

Along $u_n = \pi/2 + 2\pi n$ this is e^{-1}; along $v_n = 3\pi/2 + 2\pi n$ it is e. Both sequences tend to infinity, and the values have different limits. Therefore f/g has no limit.

Conclusion. The extended derivative expression tends to zero, but the required nonvanishing hypothesis fails. There is no contradiction to L'Hospital's rule.

31 Taylor's theorem

A Taylor polynomial matches a function's derivatives at its center. Matching those derivatives does not by itself show that the infinite series represents the function. The missing step is to show that the remainder tends to zero. With Ross's convention the polynomial in R_n stops at degree $n-1$:

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

Taylor's theorem bounds this error using the nth derivative at a point between c and x. Later exercises use Newton's method, which replaces a curve by its tangent line and uses that line's zero as the next approximation.

31.1 Exercise 31.1: The cosine series and its error

What we must show. Find the Taylor series at 0 and prove that it equals $\cos x$ everywhere.

Compute the coefficients.

Successive derivatives of cosine are $\cos x, -\sin x, -\cos x, \sin x$, then repeat. At zero, the odd derivatives vanish and the $2k$th derivative is $(-1)^k$. Thus the Taylor series is

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

Control the error, rather than assuming the series is correct.

Every derivative of cosine has absolute value at most 1. Taylor's theorem therefore gives $|R_n(x)| \leq |x|^n / n!$ for each fixed real x. This bound tends to zero: if $a_n = |x|^n / n!$, then $a_{n+1}/a_n = |x| / (n+1) \leq 1/2$ eventually (and $x = 0$ is immediate). The tail is consequently bounded by a geometric sequence tending to zero.

Conclusion. The displayed series equals $\cos x$ for every $x \in \mathbb{R}$.

31.2 Exercise 31.2: The hyperbolic sine and cosine series

What we must show. Obtain series at zero for $\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$, valid on all of $\mathbb{R}$.

Add and subtract the exponential series.

For each real x, the exponential series is absolutely convergent: $e^x = \sum_{k\geq 0} x^k/k!$ and $e^{-x} = \sum_{k\geq 0}(-1)^k x^k/k!$. Subtracting cancels the even terms; adding cancels the odd terms. Hence

$$\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}, \qquad \cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}.$$

Termwise addition here is legitimate because it is addition of convergent series at a fixed x.

Check that these really are Taylor coefficients.

Differentiation interchanges sinh and cosh. Their values at zero are 0 and 1, respectively. This gives exactly the displayed coefficients. Alternatively, on $[-M, M]$ all derivatives of either function are bounded by e^M, so Taylor's bound is $e^M |x|^n / n! \to 0$. For any given x, choose $M > |x|$; this proves validity at that x.

Conclusion. Both series represent their functions at every real number.

31.3 Exercise 31.3: Why the common derivative bound is unavailable

What we must show. Explain the theorem choice for $f(x) = \log(1 + x)$ in Example 2.

Inspect the hypothesis of Corollary 31.4.

That corollary requires a single constant bounding *all* derivatives on the interval in question. But

$$f^{(n)}(x) = (-1)^{n+1}(n-1)!(1+x)^{-n}, \qquad \left|f^{(n)}(0)\right| = (n-1)!.$$

Even at the center 0 these numbers have no common finite bound. No interval containing zero can satisfy the corollary's hypothesis.

Use the more precise remainder instead.

For evaluating at $x = 1$, Taylor's theorem gives a $y_n \in (0, 1)$ with

$$|R_n(1)| = \frac{1}{n(1+y_n)^n} \le \frac{1}{n} \longrightarrow 0.$$

The factorial growth of the derivative is canceled by the $n!$ in the remainder. The corollary's stronger hypothesis would miss that cancellation.

Conclusion. Theorem 31.3 applies directly; the common-bound hypothesis of the corollary fails.

31.4 Exercise 31.4: Smooth cutoffs and a smooth step

What we must show. Construct the four requested infinitely differentiable functions.

Plan. Start from the smooth function in Example 3,

$$\phi(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

It is important that all its derivatives at zero vanish; this is what makes the joining point smooth, despite the piecewise formula.

(a) Move the start of the positive part to a.

Set $f_a(x) = \phi(x - a)$. A translation preserves smoothness. If $x \le a$ its argument is nonpositive, so $f_a(x) = 0$; if $x > a$, its argument is positive and $f_a(x) > 0$.

(b) Reverse the direction and move the endpoint to b.

Set $g_b(x) = \phi(b - x)$. The affine change of variable preserves smoothness. This is zero for $x \ge b$ and strictly positive for $x < b$.

(c) Require both positivity conditions.

Set $h_{a,b}(x) = f_a(x)g_b(x)$. Products of smooth functions are smooth. Both factors are positive exactly when $a < x < b$. Outside this interval at least one factor is zero, so the product is zero.

(d) Turn the two cutoffs into a step.

Set

$$h^*_{a,b}(x) = \frac{f_a(x)}{f_a(x) + g_b(x)}.$$

The denominator never vanishes. Indeed, if $x \le a$ then $x < b$ and $g_b(x) > 0$; if $x \ge b$ then $x > a$ and $f_a(x) > 0$; between a and b both are positive. The permitted quotient rule for smooth functions therefore applies. For $x \le a$ the quotient is 0; for $x \ge b$ its numerator equals its denominator, so it is 1.

Conclusion. These four explicit formulas have all the required properties, including smoothness at a and b.

31.5 Exercise 31.5: A nonzero smooth function with zero Taylor series

What we must show. For $g(x) = e^{-1/x^2}$ when $x \neq 0$, $g(0) = 0$, find all derivatives at zero and compare g with its Taylor series.

(a) Recognize a smooth composition.

Using ϕ from Example 3, we have $g(x) = \phi(x^2)$ for every x. Since both ϕ and $x \mapsto x^2$ are smooth, so is g. Repeated chain and product rules show that each positive-order derivative of this composition is a finite sum of polynomial factors times $\phi^{(j)}(x^2)$, for integers $j \ge 1$. At zero every such term vanishes, because $\phi^{(j)}(0) = 0$. Also $g(0) = 0$. Thus $g^{(n)}(0) = 0$ for all $n = 0, 1, 2, \ldots$.

(b) Compare the function and the series point by point.

Every coefficient $g^{(n)}(0)/n!$ is zero, so the Taylor series is the identically zero series. It equals g at 0. For $x \neq 0$, $e^{-1/x^2} > 0$, so it cannot equal $g(x)$ there.

Conclusion. Smoothness alone does not imply agreement with a Taylor series, even arbitrarily close to its center.

31.6 Exercise 31.6: A telescoping proof of Taylor's theorem

What we must show. Verify the older proof using its auxiliary function F. Fix $x > 0$. As in the theorem's proof, define M by

$$f(x) = \sum_{k=0}^{n-1} \frac{x^k}{k!} f^{(k)}(0) + \frac{Mx^n}{n!}.$$

(a) Differentiate and cancel adjacent terms.

Write the auxiliary function as

$$F(t) = \sum_{k=0}^{n-1} \frac{(x-t)^k}{k!} f^{(k)}(t) + M\frac{(x-t)^n}{n!}, \qquad 0 \le t \le x.$$

For $k \geq 1$, the product rule gives

$$\frac{d}{dt}\left(\frac{(x-t)^k}{k!}f^{(k)}(t)\right) = -\frac{(x-t)^{k-1}}{(k-1)!}f^{(k)}(t) + \frac{(x-t)^k}{k!}f^{(k+1)}(t).$$

The negative term cancels the positive term from index $k-1$. Only the last positive term and the derivative of the M term remain:

$$F'(t) = \frac{(x-t)^{n-1}}{(n-1)!}(f^{(n)}(t) - M).$$

All needed derivatives exist on an open interval containing $[0, x]$; in particular F is continuous on the closed interval and differentiable in its interior. For $n = 1$ the formula follows directly from $F(t) = f(t) + M(x-t)$.

(b) Evaluate the endpoints.

The definition of M says $F(0) = f(x)$. At $t = x$ all positive powers of $x-t$ vanish, leaving $F(x) = f(x)$. Thus $F(0) = F(x)$.

(c) Apply Rolle's theorem.

There is $y \in (0, x)$ with $F'(y) = 0$. Since $x - y > 0$, the prefactor in the derivative formula is nonzero; hence $f^{(n)}(y) = M$. Substituting into the defining equation for M yields Taylor's remainder.

Conclusion. The auxiliary function proves the required mean-value expression for the remainder.

31.7 Exercise 31.7: Newton's method for the square root of two

What we must show. Recover the recursion from Exercise 9.5. For $f(x) = x^2 - 2$, we have $f'(x) = 2x$. Newton's update at a nonzero t_n is

$$t_{n+1} = t_n - \frac{t_n^2 - 2}{2t_n} = \frac{2t_n^2 - t_n^2 + 2}{2t_n} = \frac{t_n^2 + 2}{2t_n}.$$

With initial point $t_1 = 1$ this is exactly the earlier sequence. Every iterate is positive, because a positive t_n gives a positive numerator and denominator. Thus no zero denominator is encountered.

Conclusion. The recursion is Newton's method for $x^2 - 2 = 0$, starting at 1.

31.8 Exercise 31.8: Two real roots of a quartic

What we must show. Count the real roots of $p(x) = x^4 + x^3 - 1$ and approximate each to six decimals.

(a) Establish the shape and count crossings.

The derivative is $p'(x) = x^2(4x + 3)$. It is negative for $x < -3/4$ and positive for $x > -3/4$, except for the isolated zero at $x = 0$. Thus p is strictly decreasing on $(-\infty, -3/4]$ and strictly increasing on $[-3/4, \infty)$. For strict increase across zero, apply the Mean Value theorem separately on either side and combine the inequalities. Now $p(-2) = 7 > 0$, $p(-1) = -1 < 0$, $p(0) = -1 < 0$, and $p(1) = 1 > 0$. The Intermediate Value theorem supplies one root in $(-2, -1)$ and one in $(0, 1)$. Strict monotonicity on the two branches rules out any others.

(b) Run the explicit update.

Newton's formula is

$$x_{n+1} = x_n - \frac{x_n^4 + x_n^3 - 1}{4x_n^3 + 3x_n^2}.$$

Two runs, retaining guard digits, give

n	$x_0 = -1.5$	$x_0 = 0.8$
0	−1.500000000000	0.800000000000
1	−1.398148148148	0.819758064516
2	−1.380747193802	0.819173040661
3	−1.380277903561	0.819172513397
4	−1.380277569098	0.819172513396

As a check independent of apparent convergence, direct substitution gives opposite signs at the endpoints of each interval $(-1.380277570, -1.380277568)$ and $(0.819172512, 0.819172515)$. Each interval contains the unique root on its branch and lies wholly within the rounding range for the answer below.

Conclusion. The two roots, rounded to six decimal places, are −1.380278 and 0.819173.

31.9 Exercise 31.9: Solving the cosine fixed-point equation

What we must show. Approximate the solution of $x = \cos x$ to six decimals. Set $f(x) = x - \cos x$. Since $f'(x) = 1 + \sin x > 0$ on $(0, \pi/2)$, the root there is unique. Newton's iteration is

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}.$$

Starting at $x_0 = 1$ gives successively

$$0.750363867840, \quad 0.739112890911, \quad 0.739085133385, \quad 0.739085133215.$$

The signs of f at 0.739085132 and 0.739085135 are negative and positive, respectively. Continuity and strict increase bracket the root, so the displayed digits are not justified solely by two similar iterates.

Conclusion. The root is $x = 0.739085$ to six decimal places.

31.10 Exercise 31.10: Solving an exponential equation

What we must show. Approximate the root of $xe^x = 2$ in $(0, 1)$ to six decimals. Use $f(x) = xe^x - 2$, with $f'(x) = (1 + x)e^x > 0$ on $(0, 1)$. Thus there is at most one root there; the opposite endpoint signs give existence by continuity. Newton's update is

$$x_{n+1} = x_n - \frac{x_n e^{x_n} - 2}{(1 + x_n)e^{x_n}}.$$

Starting with $x_0 = 1$ gives

$$0.867879441171, \quad 0.852783373416, \quad 0.852605526369, \quad 0.852605502014.$$

Direct substitution gives $f(0.852605501) < 0 < f(0.852605504)$, so the unique root is in that small interval.

Conclusion. The six-decimal value is 0.852606.

31.11 Exercise 31.11: A convergent Newton sequence must approach a zero

What we must show. Under the stated bounded-derivative hypotheses, prove $f(x) = 0$ when the Newton iterates converge to the interior point x.

Rearrange the recursion before taking limits.

Since $f'(x_{n-1}) \neq 0$, the update is defined and implies

$$f(x_{n-1}) = (x_{n-1} - x_n)f'(x_{n-1}).$$

Choose $B < \infty$ with $|f'(t)| \leq B$ throughout (a, b). Convergence of (x_n) means both x_n and x_{n-1} tend to x, so their difference tends to zero. Consequently

$$|f(x_{n-1})| \leq B\,|x_{n-1} - x_n| \longrightarrow 0.$$

Identify this limit using continuity.

Differentiability at the interior point x implies continuity there, so also $f(x_{n-1}) \to f(x)$. Uniqueness of a real limit gives $f(x) = 0$. We did not assume f' was continuous or bounded away from zero; the boundedness from above was the estimate actually used.

Conclusion. Every convergent sequence of these iterates with an interior limit converges to a root.

31.12 Exercise 31.12: An interior point is no farther than an endpoint

What we must show. If $a < c < b$, prove $|c - x| \leq \max\{|a - x|\,, |b - x|\}$ for every real x.

Case 1: $x \leq c$**.**

Then $c - x \geq 0$ and $b - x > 0$, so $|c - x| = c - x \leq b - x = |b - x|$.

Case 2: $x > c$**.**

Then $x - c > 0$ and $x - a > 0$, so $|c - x| = x - c \leq x - a = |a - x|$. These cases exhaust the real line. In either case the relevant endpoint distance is at most the maximum of the two endpoint distances.

Conclusion. The claimed inequality follows for all $x \in \mathbb{R}$.